



FOR QUANTITY SURVEYORS & CONSTRUCTION PROFESSIONALS

Turn Your Data Into Decisions — Build Real Dashboards for Real Projects

Module 01: Power BI
Foundations

Module 02: Data Modelling &
DAX

Module 03: Construction
Dashboards

Module 04: Live Projects &
Sharing

4

Core
Modules

30+

Topics
Covered

10+

Real Construction
Dashboards Built



Empowering Built Environment Professionals

POWER BI FOR

QS & CONSTRUCTION



www.aclacademysl.com



+94 76 313 6718



ABOUT ACL ACADEMY

WHO WE ARE

ACL Academy is a modern training platform dedicated to developing industry-ready professionals through practical, real-world learning in construction and the built environment. Founded by a Chartered professional with extensive Middle East experience, ACL Academy bridges the gap between academic knowledge and industry practice through hands-on training, real project scenarios, and digital workflows.

OUR VISION

To become a leading global platform that transforms built environment professionals by integrating real industry practice with advanced digital and AI-driven solutions.

OUR MISSION

To equip students and professionals with practical, job-ready skills across the built environment by delivering training that reflects real project environments, enhanced with modern tools, digital workflows, and Artificial Intelligence.

WHY CHOOSE ACL ACADEMY?

Industry Experts in the Lecture Room

Every module is delivered by active construction professionals with real world experience — not academics reading from textbooks.

Practical, Project-Based Learning

You work through real documents, real BOQs, real payment applications and real contract scenarios drawn from live construction projects.

Specific Contract Knowledge

Full coverage of FIDIC, NEC and JCT contracts, procurement practices, and the commercial structures used by leading regional developers.

Interview & Career Preparation

The programme concludes with dedicated interview coaching and a live mock interview session - so you leave job-ready, not just course-complete.

RICS-Aligned Training Standards

Course content, competency framework, and assessment approach are designed to align with RICS professional standards and APC competency requirements.

Trusted by Professionals Across the Gulf

Our graduates are working as QS professionals across the UAE, Saudi Arabia, Qatar, Kuwait, Bahrain, Oman, and Sri Lanka.



MEET YOUR LECTURER

IA

IRSATH ALI

Senior Data & AI Scientist

Lecturer at NIBM & IIT Sri Lanka

12+ Years Industry Experience

Senior Data & AI Scientist

Lecturer — NIBM & IIT Sri Lanka

Machine Learning & AI Strategy

Power BI & Business Intelligence

12+

Yrs Exp

100+

Trained

AI+BI

Specialisation

ABOUT THE LECTURER

Irsath Ali is an elite Senior Data & AI Scientist and academic with over 12 years of trailblazing industry experience spanning data science, machine learning, and corporate AI strategy. As an active lecturer at top-tier institutes including NIBM and IIT Sri Lanka, Irsath has trained hundreds of professionals and students, mastering the art of breaking down highly complex systems into actionable business intelligence.

A recognised expert in Artificial Intelligence and Power BI, Irsath specialises in deploying machine learning models and creating sophisticated, data-driven dashboards that transform corporate decision-making. His sessions equip professionals with the exact tools — ranging from Python automation to advanced Power BI analytics — needed to lead the AI transformation across modern industries.

KEY AREAS OF EXPERTISE

◆ Data Science & Machine Learning

◆ Power BI Advanced Analytics

◆ Artificial Intelligence Strategy

◆ Python Automation

◆ Corporate Business Intelligence

◆ Interactive Dashboard Development

◆ Academic Lecturing (NIBM & IIT)

◆ AI-Driven Decision Making



ABOUT THIS TRAINING

Power BI for Quantity Surveyors & Construction Professionals

Your Data Is Already There - Now Make It Work for You

Every construction project generates enormous amounts of data - BOQs, payment applications, variation logs, subcontractor accounts, cost reports, programme updates, and more. But most of this data sits in Excel spreadsheets, untouched and unanalysed. Power BI transforms this raw data into live, interactive dashboards that give project teams, QS professionals, and commercial managers the real-time insight they need to make better decisions, control costs, and report with confidence.

4

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Real Construction
Dashboards Built

01

Built for Construction — Not Generic Training

Every topic, every exercise, and every dashboard in this training is built around real construction and QS data. You will not be working with sales figures or marketing metrics — you will be building cost reports, payment trackers, and project dashboards that you can take directly into your working environment.

02

No Coding. No IT Background Required.

Power BI is a visual, drag-and-drop business intelligence tool. This training assumes no prior programming knowledge. If you can use Excel, you can learn Power BI. Our step-by-step approach takes you from the basics of connecting data all the way to publishing professional-grade interactive dashboards.

03

Practical Outcome from Day One

From Module 01, you are building real dashboards. By the end of the training, you will have a complete portfolio of construction-specific Power BI reports — Cost Control Dashboard, Payment Application Tracker, BOQ Analysis, CVR Report, and Project Progress Dashboard — ready to use on your projects.



MODULE 01

POWER BI FOUNDATIONS — Getting Started with Construction Data

8 Topics

This module introduces Power BI Desktop, Service and Mobile — covering data connection, transformation, and your first construction reports from scratch. No prior experience needed.

Introduction to Power BI

- 01** Overview of Power BI Desktop, Power BI Service, and Power BI Mobile. Understanding the Power BI ecosystem and how it fits into a construction QS workflow.

Connecting to Construction Data

- 02** Connecting Power BI to Excel workbooks, CSV files, and databases that store BOQ data, payment applications, and cost reports. Live vs imported connections.

Power Query — Cleaning Your Data

- 03** Using Power Query Editor to clean, reshape, and transform messy construction data — removing duplicates, fixing data types, merging tables, and standardising column formats.

Understanding Tables & Relationships

- 04** How to structure your data as a proper data model. Creating relationships between BOQ tables, payment tables, and project tables - the foundation of every good Power BI report.

Building Your First Visuals

- 05** Creating bar charts, line charts, pie charts, tables, and KPI cards using construction data. Formatting visuals to look professional and easy to read by project stakeholders.

Filters, Slicers & Interactivity

- 06** Adding slicers to filter by project, trade, cost code, or time period. Cross-filtering between visuals and creating interactive reports that project managers can use.

Creating Your First Report Page

- 07** Designing a complete, professional report page layout. Arranging visuals, adding titles, logos, and company branding to produce a report ready to share with clients.

Publishing to Power BI Service

- 08** Uploading your report to Power BI Service (the cloud), setting up a workspace, and sharing access with your team or project stakeholders securely.



MODULE 02

DATA MODELLING & DAX — Calculations That Drive QS Decisions

8 Topics

This module covers the analytical engine behind Power BI — teaching you how to build data models and write DAX formulas specifically for QS calculations: cost variance, earned value, cumulative payments, and more.

Star Schema Data Modelling

- 01** Building a proper data model with fact tables (payments, costs, variations) and dimension tables (projects, trades, periods). Understanding filter flow and model relationships for accurate QS reporting.

Key QS Measures

- 03** Writing DAX measures for the most important QS calculations: Contract Sum, Total Certified, Amount Remaining, Retention Held, Variation Total, and Net Payment Due.

Cumulative & Time Intelligence

- 05** Using DATEADD, TOTALYTD, and CALCULATE to build cumulative S-curves, monthly spend analysis, and period-over-period cost comparisons for progress reports.

Subcontractor Account Measures

- 07** Creating measures for subcontractor back-to-back reconciliation - subcontract value, certified to date, retention withheld, and amount owed - essential for main contractor QS teams.

DAX Fundamentals

- 02** Introduction to Data Analysis Expressions (DAX) - the formula language of Power BI. Understanding row context, filter context, and how DAX differs from Excel formulas for construction data.

Cost Variance & Earned Value

- 04** Calculating Budget vs Actual cost variance, Earned Value (EV), Cost Performance Index (CPI), and Schedule Performance Index (SPI) - critical metrics for project commercial control.

Payment & Cash Flow Calculations

- 06** Building measures to track cumulative certified amounts, outstanding payments, retention release schedules, and projected cash flow across multiple projects or periods.

Dynamic Reporting with Parameters

- 08** Using What-If parameters and field parameters to create dynamic reports where stakeholders can switch between project, trade, or time period views without changing the underlying data.

This is the centrepiece of the training. Every dashboard is built from scratch using real construction project data — you leave with a complete portfolio of working Power BI reports ready to use in your QS or project management role.

Cost Control Dashboard

- 01** Live tracking of budget vs actual spend by trade, cost code, and project. Cost variance indicators, remaining budget alerts, and trend lines - the single most important report for any QS.

Payment Application Tracker

- 02** Tracking all interim payment applications - amounts submitted, certified, paid, and outstanding. Cumulative S-curve of certified value. Retention held and projected release schedule.

BOQ Analysis Dashboard

- 03** Visualising Bills of Quantities data - value by trade, element, or cost category. Comparing tendered rates, measured quantities, and final account values across the project scope.

Cost Value Reconciliation (CVR)

- 04** The main contractor's most critical commercial report - comparing value certified (income) against costs incurred to calculate profit/loss, margin, and project financial health at any point in time.

Variation & Claims Log

- 05** A live variation register tracking all instructed and potential variations - claimed value, agreed value, status, and open items. Flags unresolved variations and tracks progress to final agreement.

Subcontractor Account Dashboard

- 06** Complete subcontractor commercial management dashboard - subcontract value, certified to date, retention, net payments, and back-to-back reconciliation against main contract income.

Project Progress Dashboard

- 07** Planned vs Actual programme and cost S-curves. Earned value analysis. Milestone tracking and delay indicators - giving clients and project teams full visibility of schedule and cost performance.



MODULE 04

ADVANCED FEATURES, LIVE PROJECTS & SHARING YOUR WORK

7 Topics

The final module covers advanced Power BI features, live data refresh, sharing with project teams, and a complete live project exercise — putting everything together into a professional, shareable construction analytics report.

Advanced Visualisations

- 01** Drill-through reports, drill-down hierarchies, decomposition trees, and custom tooltips. Creating dynamic report navigation and page-level filtering for project-level deep dives.

Conditional Formatting & KPIs

- 02** Using conditional formatting to highlight over-budget items, overdue payments, and critical variations in red. Building KPI cards with targets and status indicators for executive dashboards.

Scheduled Refresh & Live Data

- 03** Setting up automatic data refresh from Excel, SharePoint, or SQL so your Power BI dashboards always reflect the latest project data without manual updates.

Sharing, Workspaces & Security

- 04** Publishing reports to Power BI Service, organising dashboards into workspaces, and controlling who can view, edit, or export your construction reports, essential for multi-party projects.

Power BI Mobile

- 05** Optimising dashboards for mobile view so project managers and directors can access live project data from site on any device.

Integrating with Excel & SharePoint

- 06** Connecting Power BI to Excel models and SharePoint lists that construction teams already use, seamlessly extending existing workflows.

Live Project Capstone Exercise

- 07** A complete end-to-end project given a real construction dataset, you build, format, and publish a full suite of professional dashboards covering cost, payment, variations, and progress.

START TURNING DATA INTO DECISIONS

By the end of this training, you will have built 7+ working Power BI dashboards using real construction data — and the skills to build any report your project or employer needs.



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